

Improve Generalization

Project: Responsible Plant Classifier

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Overfitting

Without opening the old lesson, explain one key idea from Overfitting. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Train, Validation, Test

Without opening the old lesson, explain one key idea from Train, Validation, Test. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Interpret a gap for Overfitting and model capacity. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Overfitting and model capacity. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Changed independent case

Spotted leaves have blue backgrounds and healthy leaves white backgrounds in training. Design a generalisation investigation with fresh evaluation.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

All spotted training leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

6. Solve the given case

A classifier relies on tray colour; new trays reduce performance.

Propose controlled improvement.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A team proposes more representative training examples and a simpler model. How should it compare changes while protecting the final test?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 30 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Improve Plant Classifier with one approved change and a full before/after evidence table.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

All spotted training leaves use black backgrounds. Propose a collection and comparison that tests the background shortcut without tuning on final test examples.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Select from evidence

Choose among the three worked candidates and explain the training trap.

My reasoning and evidence

13. W2 Interpret a gap

Does a training-validation gap alone prove overfitting?

My reasoning and evidence

14. W3 Propose a controlled change

Describe one fair next development comparison.

My reasoning and evidence

15. W4 Apply independently

Models A training 4 validation 5, B training 1 validation 2, C training 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

16. W5 Limit a diagnosis

What extra evidence would you inspect before concluding a model memorized noise?

My reasoning and evidence

17. Distribution shift — W1 Compare conditions

Calculate each worked error rate and the difference.

My reasoning and evidence

18. Distribution shift — W2 Read pooled evidence

Why is 8/40 incomplete by itself?

My reasoning and evidence

19. Distribution shift — W3 Wait for labels

Can changed predictions alone prove an accuracy drop?

My reasoning and evidence

20. Distribution shift — W4 Apply independently

Old-like data has 3 errors/30 and new-condition data 5/10. Give both rates and the pooled rate.

My reasoning and evidence

21. Distribution shift — W5 Plan a response

State a safe classroom response to inputs outside the supported range.

My reasoning and evidence

22. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

23. Pictorial case practice: Improve Generalization

Test whether a plant model follows the leaf or a convenient background shortcut.

Toy training photos show spotted leaves on red cards and healthy leaves on grey cards. Intended use allows both backgrounds for either class.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Improve Generalization

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Notice correlation

Background colour perfectly tracks the labels in this training sample.



2. Design crossed cases

Collect spotted/red, spotted/grey, healthy/red, healthy/grey with plant grouping preserved.



3. Inspect fixed model

Compare predictions for the same leaf on both backgrounds; do not assume an error cause before observing it.

CHANGE ONE THING

The unchanged model predicts spotted for both red-background classes and healthy for both grey-background classes.

Explain the changed result using the matching lesson.

What fresh evidence is needed after adding crossed training data? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.